(a) layered architecture — the LLM proposes raw inputs; deterministic calculators compute; the verifier re-proves

1	L1 input boundary	goal prose -> DesignInput (raw only, extra="forbid"); _reject_derived_fields rejects any derived key — a typed number is not trusted
2	L2 agent brain	LLMClient (deepseek-v4-flash / pro, T=0.2, thinking off, any OpenAI-compatible endpoint) + DesignAgent ReAct loop (12 iters, 8 tool calls/turn)
3	L3 tool registry	46 calculators + submit_design gate, 10 classes; pydantic params -> JSON Schema; each tool carries a worked example + common unit trap
4	L4 calculator core	10 pure calc/ modules (microfluidics · cell · culture · spheroid · o2 · barrier · dosing · pk · stats · units) + pint; derived numbers are produced ONLY here
5	L5 domain blocks	blocks.py: 7 declarative Blocks (flow/cells/culture/spheroid/dosing/stats/pk) -> ALL_* tables of keys, sanity bands, canonical units; import-time validation
6	L6 verifier	5 layers re-derive every derived number from the agent's own inputs: arithmetic -> units -> sanity -> safety -> prose numbers
7	L7 knowledge base	physiology.py: 10 cell profiles (shear · TEER · O2 · BSL, with sources) anchored into the prompt; plate-format & spheroid-format tables
8	L8 outputs	DesignPlan JSON · Markdown SOP · provenance/ELN (formula + inputs + units + code version; json | csv) · Gradio UI · CLI

(b) bounded agentic workflow — one goal in, a design whose every number was computed by calc/ and re-proved by verify/ comes out

1. goal
wet-lab goal in natural language + assumptions

2. ReAct tool loop
LLM proposes raw inputs; calls calculators to reason; prose answers are refused

3. submit_design
hard gate: derived fields rejected; DesignInput extra=forbid

4. verifier re-proves
ok -> accept · review_required -> fix raw inputs only

5. SOP + design JSON
verified protocol + provenance / ELN export

review_required / validation_error -> the agent fixes ONLY the raw inputs it proposed (never hand-writes a derived number)

benchmark systems — same tools · same loop · same scoring: bare · soft-gate · self-verify · tool_no_gate · finetuned-ext · labwright

(c) verifier — re-derives every derived number from the agent's own raw inputs

1 arithmetic
re-runs every governing equation
(checker.py, 8 domains)

2 units & dimensions
17 alias traps via pint
dyn/cm2-as-Pa = 10x, mL/min-as-uL/min...

3 physiological sanity
soft / hard bands
shear 1e-3-10 Pa soft, 1e-4-50 hard

4 safety & compliance
SafetyConfig: DMSO <0.5%, dose caps, vehicle control, BSL hints, institution

5 prose-number gate
numbers written in the narrative must match a design value (thresholds skipped)

a design is accepted only when no layer reports an error · prose assertions are warnings, so an honest design is never blocked

(d) calculator toolbox — 46 tools in 10 classes (pure functions · pydantic-schema'd · pint units · worked example + unit traps)

microfluidics · 6
wall_shear_stress · flow_rate_for_shear_stress · reynolds_number · pressure_drop · residence_time · o2_delivery_rate

cell · 11
seeding_cell_count · cell_count_after_time · time_to_confluence · cells_per_well · medium_volume_per_well · hemocytometer_count · trypan_blue_viability · confluence_to_cell_count · cell_count_to_confluence · passage_split_ratio · moi_virus_volume

dosing · 3
molarity_from_mass · dilution_volume · dmso_fraction

stats · 2
sample_size_per_group · technical_replicates

published · 1
verify_published_protocol

3d_culture · 5
spheroid_volume · spheroid_diameter_from_cells · cells_per_spheroid_for_diameter · medium_volume_per_spheroid · spheroids_from_suspension

physiology · 1
cell_physiology

o2 · 6
o2_penetration_depth · spheroid_necrotic_fraction · o2_supply_vs_demand · o2_peclet · o2_damkohler · o2_po2_conversion

barrier · 6
teer_from_resistance · transendothelial_resistance · papp_from_flux · flux_from_papp · clearance_from_papp · effective_permeability

pk · 5
extraction_ratio · clearance_uLmin · half_life_h · accumulation_ratio · mass_cleared_ug_h

calc/ — 10 pure modules; derived numbers are produced HERE and only here: microfluidics · cell · culture · spheroid · o2 · barrier · dosing · pk · stats · units

$\tau = 6\mu Q/(wh^2)$ · $Re = \rho\bar{u}D\_h/\mu$ · $\Delta P = 12\mu QL/(wh^3)$ · $n = 2(z\alpha+z\beta)^2\sigma^2/\delta^2$ · $E = 1 - Cout/Cin$ · $t_{1/2} = \ln 2 \cdot V/Cl$

$\delta = \sqrt{2 \cdot D \cdot C0/q}$ · $TEER = (R - Rblank) \cdot A$ · $Papp = J/(A \cdot C0)$ · $C = N \cdot D \cdot 1e4$

every derived field pinned to one canonical unit by pint (dyn = 1e-5 N); a new domain = one calc/ module + one register_tool call — not a fork

(e) internal components · benchmark systems · honest boundary

the agent & brain

DesignAgent — ReAct loop; ends only on submit_design; prose refused; on review_required fixes raw inputs only

LLMClient — OpenAI-compatible; deepseek-v4-flash/pro; T=0.2; thinking off; retry 3x backoff; max_tokens 8192

tool_no_gate ablation — same calculators & loop, verifier off; isolates what the verification layer adds

cell_physiology — read-only lookup over the 10-profile physiology registry

verify_published_protocol — same calculators run backwards over a paper's own inputs (consistent / discrepancy / …)

UI (Gradio two-axis badge) · CLI · Colab — all entry points into the same verified pipeline

fine-tuned extractor (single-pass, not an agent)

Qwen2.5-1.5B-Instruct + LoRA (r=16) tuned on goal -> raw-inputs; goes through the SAME gate (reject derived -> DesignInput -> build_design -> verify)

committed eval: JSON parse 1.0 · extract->verify 0.9976 vs untuned 0.40

the agent and the extractor are the two independent paths that produce raw inputs; both feed the identical calculators and verifier

gold sets: 24-reading · 15-blind (8 cold + 5 prompt-backed + 2 scenario) · 15-3D-spheroid · 14-plate-culture · 14-perfused-PK

metrics: usable · self-consistent · hallucination · failure reason · unit-misread · target-selection · cold-split

honest boundary & committed results

the verifier checks arithmetic & internal consistency, not target selection — the UI marks both (green = arithmetic verified / amber = targets model-proposed)
reading set (24): usable 88% flash / 100% pro; hallucination 0.125 / 0.000
5-seed Wilson 0.925 [0.864, 0.960] / 0.958 [0.906, 0.982]
blind set (15): usable 40% / 47% (cold-only 38%); hallucination 0.000 (attack-tested) · self-consistency 100%

hallucination == 0.000 is attack-tested (prose refusal · derived-field rejection · tamper detection · prose assertions), not "zero by construction"